Question			Answer	Marks	AOs		Guidance
13	(a)		eg. Neglect air resistance Constant gravity Projectile is a particle	B1 [1]	1.2	One sensible statement.	Do not accept level ground
13	(b)		$u_x = 35\cos\theta$ giving $x = (35\cos\theta)t$	B1	3.3	Award seen in any form	
			$u_y = 35\sin\theta$ giving $y = (35\sin\theta)t - \frac{1}{2}gt^2$	B1	3.3	soi	
			Substitute for t	M1	3.3	Substituting for t in their equation for y	
			$y = (35\sin\theta)\left(\frac{x}{35\cos\theta}\right) - \frac{1}{2}g\left(\frac{x}{35\cos\theta}\right)^2$	A1	1.1	Award in any form ISW	
			$\left[y = x\tan\theta - \frac{x^2}{250\cos^2\theta}\right]$	[4]			
13	(c)		EITHER Using $s = 22.5$, $u = 35\sin\theta$, $v = 0$, $a = -9.8$ $v^2 = u^2 + 2as$	M1	3.1b	Using any *suvat* in y-direction with $v_y = 0$	
			$0 = (35\sin\theta)^2 - 2\times 9.8\times 22.5$	A1	1.1a	Correct equation for θ only	Either $\theta = 37°$ or $\cos\theta = \frac{4}{5}$, $\tan\theta = \frac{3}{4}$ may be used.
			$\sin\theta = 0.6$ [giving $\theta = 36.9°$]	A1	1.1	allow 37º	
			Use the trajectory with $x = 110$ $y = 110\times\tan\theta° - \frac{1}{250\cos^2\theta°}\times 110^2$	M1	3.1b	Allow in terms of θ FT their value for θ if used	Allow this M mark for $t = \frac{110}{35\cos\theta} = \frac{55}{14}$ used in a suitable equation for y
			$= 6.875$ So it goes over the wall	A1 E1 [6]	1.1 3.2a	Correct y value Conclusion in context from correct values	

Question	Answer	Marks	AOs		Guidance
	OR Using $s = 22.5, \quad u = 35\sin\theta, \quad v = 0, \quad a = -9.8$ $v^2 = u^2 + 2as$ $0 = (35\sin\theta)^2 - 2\times 9.8\times 22.5$ $\sin\theta = 0.6$ giving $\theta = 36.9°$	**M1** **A1** **A1**		Using any *suvat* in y-direction with $v_y = 0$ Correct equation for θ only allow 37º	Either $\theta = 37°$ or $\cos\theta = \frac{4}{5}$, $\tan\theta = \frac{3}{4}$ may be seen.
	Use the trajectory with $y = 5$ $5 = x\tan\theta° - \frac{1}{2450} g\sec^2\theta° x^2$ $\frac{1}{160}x^2 - \frac{3}{4}x + 5 = 0$ $x = 7.089, \quad 112.9$	**M1** **A1**		Allow in terms of θ FT their value for θ if used Both correct x-values	Allow this M mark for roots of their $y = 5$ used in a suitable equation for x.
	So particle is above the height of the wall between 7 m and 112.9 m away, so when $x = 110$ m so it does not hit the wall	**E1** **[6]**		Conclusion in context from correct values	